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ScienceDirect
Energy Reports 8 (2022) 313–321
www.elsevier.com/locate/egyr

2021 8th International Conference on Power and Energy Systems Engineering (CPESE 2021), 10–12 September 2021, Fukuoka, Japan

Dielectric analysis of degraded stator bars of a hydro generator

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Received 1 November 2021; accepted 8 November 2021
Available online 26 November 2021

Abstract

This paper presents the investigation of dielectric analysis of degraded stator bars by Polarization and Depolarization Current (PDC) measurement. The stator bars obtained from the 30-year in-service synchronous machine were divided into two groups. The first group was the unmodified stator bars. The second group was the simulated degradation stator bars which were composed of the thermal stress stator bars, conductive tap abrasion stator bars, multiple air cavity stator bars, multiple oil filled stator bars and multiple water filled stator bars. PDC test experiment was performed with the stator bar samples. PDC test results were analyzed in time domain and in frequency domain with the derived parameters i.e. R_{dc}, real and imaginary part of complex capacitance, C ratio, conduction loss factor and polarization loss factor. It was found that PDC shapes obtained from the conductive tape abrasion stator bar and multiple water filled cavity stator bar were clearly different from that of the unmodified stator bar. For other case studies, the derived parameters showed the usefulness in analysis the degraded stator bar characteristics. For example, the increase in the imaginary capacitance and conduction loss factor can be observed from most cases of degraded stator bar test results.
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Peer-review under responsibility of the scientific committee of the 8th International Conference on Power and Energy Systems Engineering, CPESE, 2021.

Keywords: Polarization and depolarization current; Rdc; C ratio; Polarization loss factor; Conduction loss factor

1. Introduction

Hydro generators play an important role in electrical generation plants, industrial sectors and etc. Since they are capital assets, and their failure may result in immense financial losses. Hence, they need a reliable maintenance plan. From the survey of CIGRE study committee SC11, EG11.02 in [1,2], it was reported that various kinds of damages, i.e., bearing damage, thermal damage, mechanical damage, and insulation damage caused the machines to fail. Focusing on the insulation problem, the aged or gradually deteriorated insulation due to the thermal, electrical,

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https://doi.org/10.1016/j.egyr.2021.11.143
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Table 1. Labeled test object and parameters in time domain and frequency domain.

	Label	C_{geo}(pC)	R_{dc}(TΩ)	PI at 60 & 600 s	C_{ratio} at 0.00001 Hz	C_{ratio} at 0.0005 Hz	tan δ_σ at 0.0003 Hz	tan δ_σ at 0.3 Hz	tan δ_{pol} at 0.0003 Hz	tan δ_{pol} at 0.3 Hz
The unmodified 30-year in-service stator bar	A	373	4.5	2.78	2.01	1.45	0.21	3.07×10^{-4}	0.1702	0.0457
Multi abrasion on the conductive layer stator bar	B	338	0.21	1.1	2.73	1.59	4.46	7.38×10^{-3}	0.2557	0.0583
Multiple air cavity stator bar	C	317	2.95	2.17	2.45	1.68	0.32	5.32×10^{-4}	0.1883	0.0922
Multiple mineral oil-filled cavity stator bar	D	308	4.1	4.02	1.91	1.71	1×10^{-6}	2.40×10^{-6}	1.7713	9.907
Multiple water-filled cavity stator bar	E	2440	6.4×10^{-4}	1.15	328.5	22.27	138.97	0.31	0.2205	0.1512

ambient, and mechanical stresses during operation tends to be a vital problem of HV rotating machine failure after periods of operation. Table 1 illustrates the main causes of insulation damages for HV rotating machines.

In 2009, a CIGRE working group published about hydro generator failures [3]. It was reported that 57% of the failures were attributed to stator insulation problems. This confirmed that the stator winding insulation condition needs to be carefully monitored and properly maintenance whereas the rotor insulation causes a few problems of hydro generator failure.

There are various standard methods applied for insulation testing to verify the insulation condition, such as the testing of insulation resistance (IR), polarization index (PI), DC high potential, power factor tip-up, and partial discharge [4–12]. For the past few decades, polarization and depolarization current (PDC) measurement has been introduced as one of the time domain dielectric spectroscopy techniques applied for insulation condition testing [13–17]. PDC test technique shows the advantage that it neither makes damages nor increases aging to tested object due to testing with a given low voltage. Currently, PDC measurement is accepted as one of the non-destructive diagnostic tests that can apply for testing various kinds of insulation of high voltage equipment [18–20]. Besides, the application of PDC for diagnosis of the insulation system of rotating machines has been interesting to perform as reported in [21–23]. Dielectric loss is one of the main factors to evaluate the integrity of the insulation. Dielectric loss consists of conduction loss, polarization loss, and ionization loss. Practically, PDC measurement provides a total loss factor or so-called dielectric dissipation factor containing both conduction loss factor and polarization loss factor. The ionization loss factor should not exist since a low step voltage was used for testing. To enhance the understanding about insulation characteristics, Fourier transform is used to transform parameters in time domain into frequency domain. Then, the derived parameters in the frequency domain, such as complex capacitance, polarization loss factor, and conduction loss factor, can be used for further analysis of the insulation condition. This paper presents the application of derived parameters for dielectric analysis of the simulated degraded stator bars caused by abrasion problem and contaminant problems compared with that of the unmodified stator bar. It is found that the derived parameters are obviously useful to discriminate the mentioned stator bar problems out of the healthy stator bar.

2. PDC measurement and analysis

2.1. Principle of PDC measurement

In PDC measurement, when switch 1 is turned on a given step DC voltage is applied to the test object. The polarization current (or the charging current) is measured and recorded for a specific time. Then, switch 2 is turned on to short circuit the test object. Simultaneously, the depolarization current (or the discharge current) is measured and recorded for a given time. The principle of polarization and depolarization measurement is illustrated in Fig. 1.

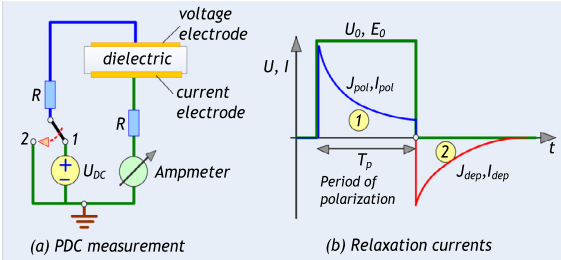

Fig. 1. Principle of polarization and depolarization current measurements.

2.2. Dielectric response in frequency domain

In this part, Fourier transform is used for transforming parameters obtained from PDC measurement in time domain into frequency domain.

The relation between a measured current $\underline{I}(\omega)$ and applied voltage $\underline{U}(\omega)$ as

$$\underline{I}(\omega) = j\omega\underline{C}(\omega)\,\underline{U}(\omega) \tag{1}$$

where $C(\omega)$ is the complex capacitance

$$\underline{C}(\omega) = C'(\omega) - jC''(\omega) \tag{2}$$

where $C'(\omega)$ and $C''(\omega)$ represents the real and imaginary parts of complex capacitance, which can be expressed in (3) and (4), respectively.

$$C'(\omega) = C_\infty + \sum_{i=1}^{n} \frac{C_i}{1 + (\omega R_i C_i)^2} \tag{3}$$

and

$$C''(\omega) = \frac{1}{\omega R_{dc}} + \sum_{i=1}^{n} \frac{\omega R_i C_i^2}{1 + (\omega R_i C_i)^2} \tag{4}$$

where C_∞ is a capacitance effective at the time instant at which the current are available, R_{dc} is a DC resistance of the test object. R_i and C_i is partial resistance and partial capacitance obtained from the curve fitting technique.

The total loss factor ($\tan\delta$), the conduction loss factor ($\tan\delta_\sigma$), and the polarization loss factor ($\tan\delta_{pol}$) can be expressed in (5)–(7)

$$\tan\delta(\omega) = \tan\delta_\sigma(\omega) + \tan\delta_{pol}(\omega) = \frac{C''(\omega)}{C'(\omega)} = \frac{\frac{1}{\omega R_{dc}} + \sum_{i=1}^{n} \frac{\omega R_i C_i^2}{1+(\omega R_i C_i)^2}}{C'(\omega)} \tag{5}$$

The conduction loss factor can be expressed as

$$\tan\delta_\sigma(\omega) = \frac{1}{\omega R_{dc} C'(\omega)} \tag{6}$$

and the polarization loss can be expressed as

$$\tan\delta_{pol}(\omega) = \frac{1}{C'(\omega)} \sum_{i=1}^{n} \frac{\omega R_i C_i^2}{1 + (\omega R_i C_i)^2} \tag{7}$$

3. Experimental detail

3.1. Preparation

The stator bars, class B insulation with 130 °C limited operating temperature, used for the experiment were obtained from a synchronous generator rated of 13.8 kV, 96.6 MVA, which has been operated for 30 years in

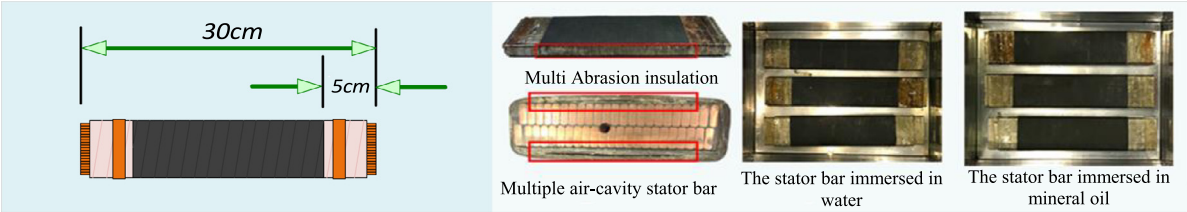

Fig. 2. Simulated degradation stator bars for the experiment.

hydropower plant as shown in Fig. 2. The slot portion of the stator bars was prepared as a test sample with a length of 30 cm. The conductive tape at both ends of the test sample was removed and then replaced with the guard-rings to avoid the effect of surface leakage current from the HV electrode when the experiment was performed.

Fifteen bar samples were prepared for this investigation. These bars were divided into two groups. The first group was the unmodified 30-year in-service stator bars. The second group was composed of the stator bars which were modified by simulating various types of stator bar degradation as shown in Fig. 2 and described below. Three stator bars were prepared for this case study.

(1) Multi abrasion on the conductive layer stator bar

The conductive tape on both sides of three stator bars was abraded to simulate the degradation.

(2) Multiple air-cavity stator bar

Four air-cavities with the dimension of $0.1 \times 5 \times 14$ cm^3 were simulated on the stator bar samples.

(3) Multiple mineral oil-filled cavity stator bar

The stator bars having four cavities were immersed in the mineral oil tray. Therefore, mineral oil went into the cavities and also coated the surface of the stator bars.

(4) Multiple water-filled cavity stator bar

The stator bars having four cavities were immersed in the water tray. Therefore, water went into the cavities and also coated the surface of the stator bars.

3.2. PDC test set-up

The PDC measurement circuit used to investigate the dielectric response is illustrated in Fig. 3(a). To perform the experiment, when switch 1 was turned on, a step DC voltage was applied to the tested specimen. Simultaneously, the polarization current was measured and recorded for a specific time. After that, switch 2 was turned on (switch 1 was turned off), the DC supplied voltage was disconnected from the test circuit, and the test object was shorted circuit, the depolarization current was measured for a given time. The PDC test results were plotted on a log–log scale. In this experiment, PDC-Analyser-1Mod was employed. Test arrangement was set up as illustrated in Fig. 3(b).

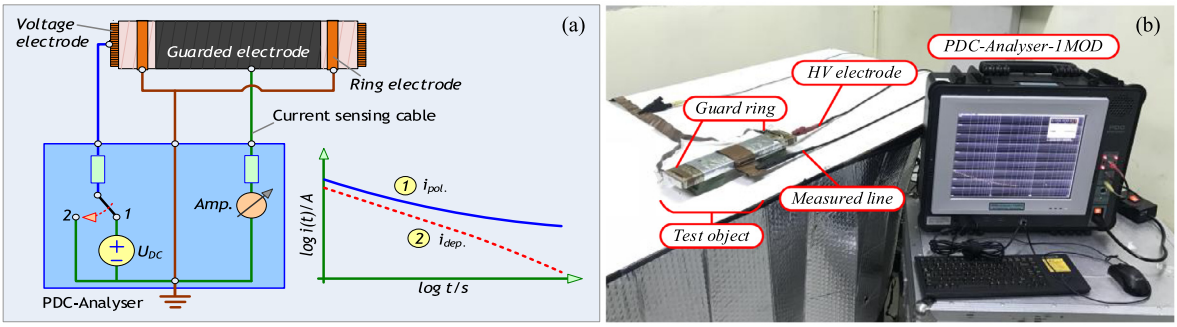

Fig. 3. (a) Test circuit diagram for PDC measurement and (b) PDC test arrangement.

3.3. PDC test procedure

The 3-step test procedure for PDC measurement was as follows: initial measurement, control measurement, and main measurement. The initial measurement was performed to determine the amount of remaining charge inside the insulation. Then, the control measurement was used to check whether there were no abnormalities occurring in the PDC test circuit or not. Finally, in the main measurement process, whereby a step DC test voltage of 500 V was applied to the test specimen for 2000 s to record the $i_{pol}(t)$ measurements, after that, the test specimen was short-circuited in order to measure $i_{dep}(t)$. Ambient temperature and humidity were also recorded during conducting the experiment. For each study case, three identical stator bars were tested to confirm the measured test data.

4. Test results and discussion

From the experiment, PDC test results in time domain were recorded and analyzed. PDC shape showed polarization and conduction mechanisms of the test object under a tested step dc voltage. In addition, polarization index (PI), C_{geo}, R_{dc}, were calculated and presented in the time domain, whereas complex capacitance, conduction loss factor, and polarization loss factor were computed in the frequency domain. From the calculation, the frequency range was divided into four parts. Very low frequency (VLF) range was defined from 0.00001 Hz to 0.001 Hz. The low frequency (LF) range was defined from 0.001 Hz to 0.1 Hz. The medium frequency (MF) range was defined from 0.1 Hz to 1 Hz. The high frequency (HF) range was defined from 1 Hz to 100 Hz. The extrapolation technique was applied to the test results to represent the analytic parameters in very low frequency range. For other frequency ranges, the analytic parameters were directly converted from the measurement in time domain. The description of the mentioned parameters is detailed below.

- The PI criteria should be more than 2 for a good condition of the tested insulation.
- The geometrical capacitance C_{geo} or high frequency capacitance represents the insulation characteristic due to the very fast polarization process.
- DC resistance R_{dc} is caused by the inherent conductivity of the insulation estimated from the difference between the polarization current and depolarization current.
- C_{ratio} is the ratio between the real capacitance and the geometrical capacitance.
- Loss factor (tan δ) consists of polarization loss factor (tan δ_{pol}) and conduction loss factor (tan δ_{σ}).

4.1. Conductive tape abrasion stator bar

The polarization and depolarization currents of the 30-year in-service stator bar compared with those of the conductive tape abrasion stator bar are illustrated in Fig. 4(a). It can be seen that the polarization and depolarization currents of the conductive tape abrasion stator bar are obviously higher than those of the 30-year in-service stator bar. Increasing the polarization current of the conductive tape abrasion stator bar causes increasing in the imaginary capacitance and conduction loss factor as well as decreasing in R_{dc} and PI, as shown in Figs. 5(a) and 6(a) and Table 1, respectively.

4.2. Multiple air cavity stator bar

The polarization and depolarization currents of the 30-year in-service stator bar compared with those of the multiple air cavity stator bar are illustrated in Fig. 4(b). The cavities or gaps between insulation layers of the simulated degradation bar increase the absorption current in the initial and last stages of the measured polarization and depolarization current curve. The imaginary part of the complex capacitance and the conduction loss factor increase a little for all frequency range, whereas the real part of the complex capacitance and polarization loss factor obviously increase only in the frequency range from 0.01 Hz to 100 Hz as represented in Figs. 5(b) and 6(b) When compared with the 30-year in-service stator bar, decrease in C_{geo} and in PI are observed for the multiple air cavity stator bar as summarized in Table 1.

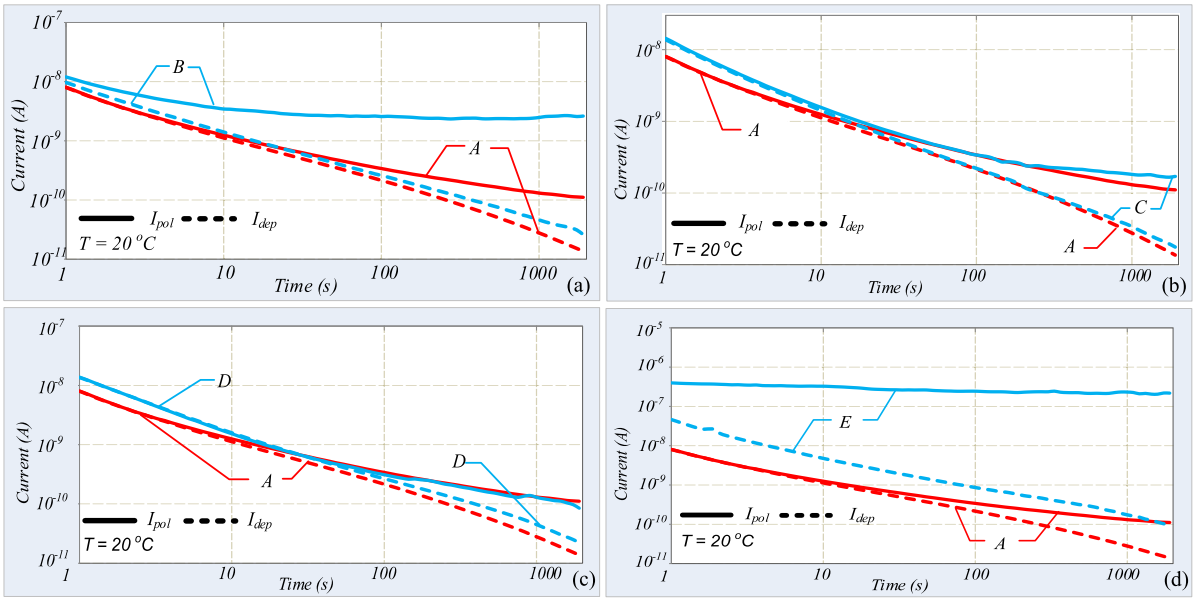

Fig. 4. The PDC of the 30-year in-service bar and (a) multi abrasion on the conductive layer stator bar, (b) multiple air-cavity stator bar, (c) multiple mineral oil-filled cavity stator bar, and (d) multiple water-filled cavity stator bar.

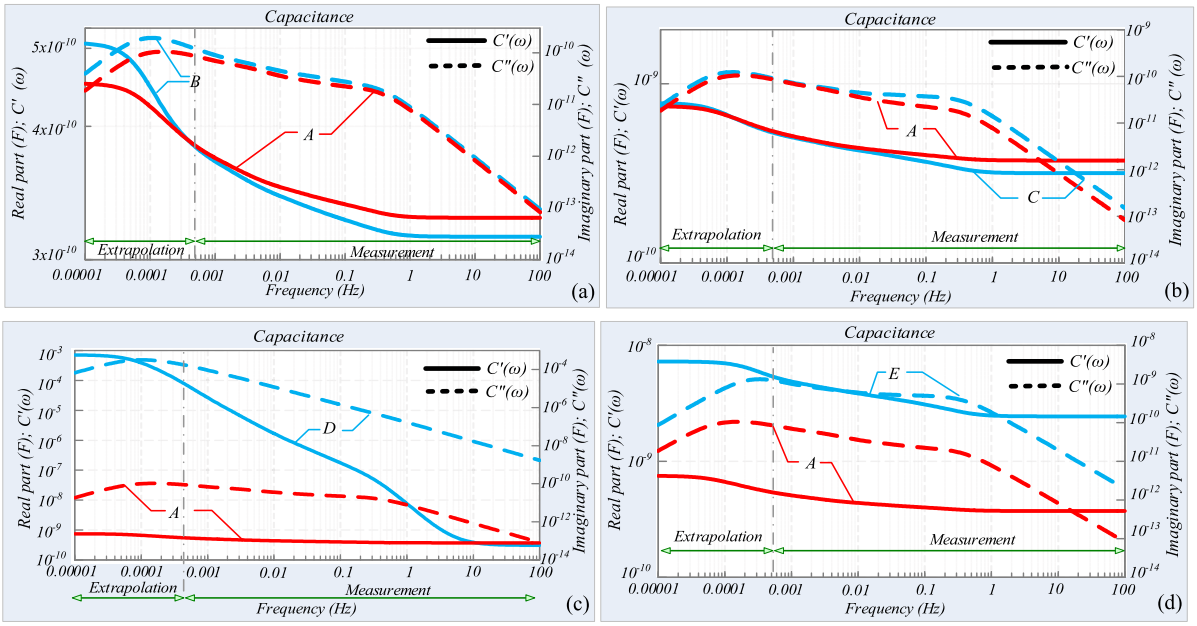

Fig. 5. Complex capacitance of the 30-year in-service bar and (a) multi abrasion on the conductive layer stator bar, (b) multiple air-cavity stator bar, (c) multiple mineral oil-filled cavity stator bar, and (d) multiple water-filled cavity stator bar.

4.3. Multiple mineral oil-filled cavity stator bar

The comparison of polarization and depolarization currents of the 30-year in-service stator bar compared with those of the multiple mineral oil-filled cavity stator bars are illustrated in Fig. 4(c). Fig. 5(c) represents the calculated

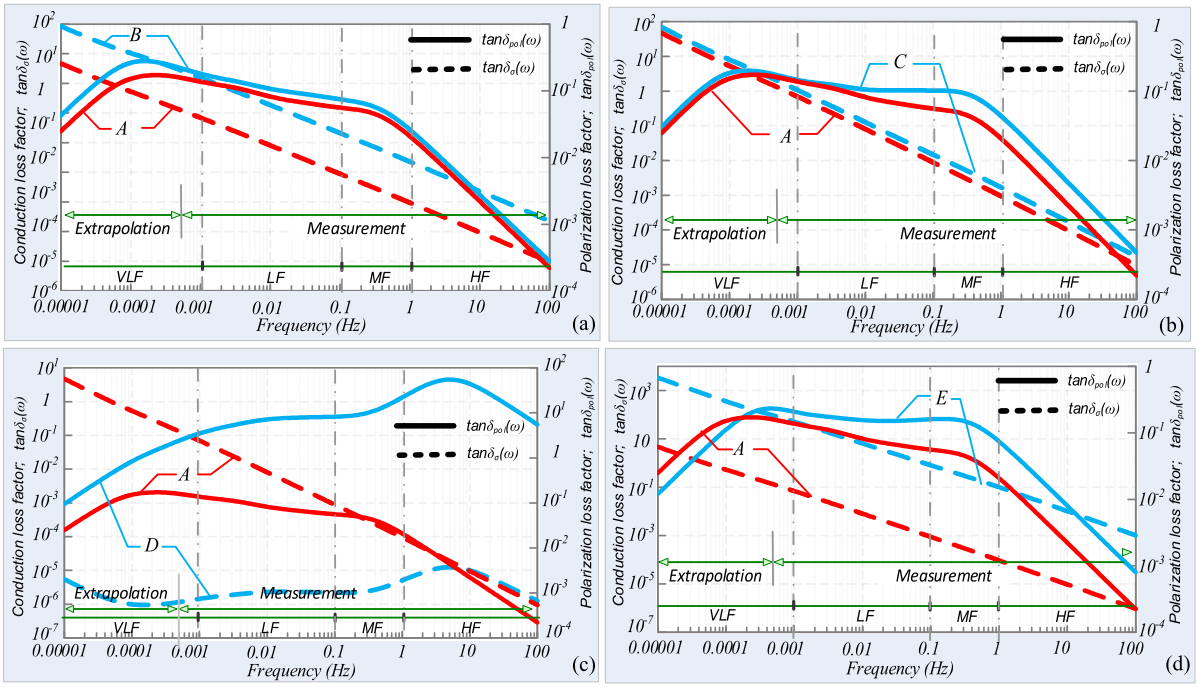

Fig. 6. The polarization and conduction loss factor of the 30-year in-service bar and (a) multi abrasion on the conductive layer stator bar, (b) multiple air-cavity stator bar, (c) multiple mineral oil-filled cavity stator bar, and (d) multiple water-filled cavity stator bar.

real part and imaginary part of the complex capacitance of the 30-year in-service stator bar compared with those of the multiple mineral oil-filled cavity stator bar. Fig. 6(c) represents the calculated polarization loss factor and conduction loss factor of the 30-year in-service stator bar compared with those of the multiple mineral oil-filled cavity stator.

From Figs. 5(c) and 6(c), the multiple mineral oil-filled cavity stator bar causes the absorption current, real and imaginary part of the complex capacitance, and the polarization loss factor increase including polarization index (see Table 1), whereas the conduction loss factor decreases compared with these of the 30-year in-service stator bar.

4.4. Multiple water-filled cavity stator bar

The comparison of polarization and depolarization currents of the 30-year in-service stator bar compared with those of the multiple water-filled cavity stator bar is illustrated in Fig. 4(d). The PDC curves of the multiple water-filled cavity stator bar show extremely high conduction current and high absorption current, which ensues from the effect of water. Fig. 5(d) represents the calculated real part and imaginary part of the complex capacitance of the 30-year in-service stator bar compared with those of the multiple water-filled cavity stator bar. Fig. 6(d) represents the calculated polarization loss factor and conduction loss factor of the 30-year in-service stator bar compared with those of the multiple water-filled cavity stator bar. It is found that the multiple water-filled cavity stator bar causes an increase in the real and imaginary parts of the complex capacitance. Besides, the conduction loss factor has a soaring increase in all frequency range different from the polarization loss factor, which only eminently raises in high frequency. From Table 1, when compared with that of the 30 year in-service stator bar, highly increase in C_{geo} and C_{ratio} but deeply decrease in R_{dc} can be observed from the data of the multiple water-filled cavity stator bar.

From Figs. 4–6, it is clear that PDC test results provide PDC shapes that are composed of useful information for distinguishing the stator bar problems. The accuracy of the PDC test result is mainly determined from the uncertainty of the current measurement and the test setup, which were strongly realized for the experiments. In addition to PDC measurement, frequency dielectric spectroscopy (FDS) is one of the worldwide accepted techniques that provides the dielectric response of the insulation consistent with the PDC test result if linear systems are considered [15,24].

5. Conclusion

In this paper, PDC measurement was performed with the 30-year in-service stator bar and the simulated degradation stator bars. It was found that PDC test results in time domain, especially in the shape of PDCs, are useful for preliminary analysis. However, some case studies show that the patterns of PDCs can be used effectively to distinguish the degraded stator bars from the normal condition stator bar. For example, PDC shapes obtained from the conductive tape abrasion stator bar and multiple water-filled cavity stator bar are evidently different from that of the unmodified stator bar. To deeply understand the characteristics of the insulation, the derived parameters i.e., R_{dc}, real and imaginary part of complex capacitance, C_{ratio}, conduction and polarization loss factors, can be used. From the test results, PIs of the multi abrasion on the conductive layer stator bar and the multiple water-filled cavity stator bar are lower than 2, which indicates high conduction current and conduction loss. The degraded stator bars in this research mostly show the increase in the imaginary capacitance and conduction loss factor except for the multiple mineral oil-filled cavity stator bar. Moreover, an increase in the polarization loss factor may be observed in some frequency.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by King Mongkut's Institute of Technology Ladkrabang, Thailand (grant number KREF 206301, Academic Melting Pot). Special thanks to Hydro Power Plant Vajiralongkorn Dam for stator bars.

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